# Software Requirements Specification

**for**

# BeeBuzz:App for Shippers and TruckDrivers

# Version 1.0 approved

**Prepared by <author>**

**VIT-AP**

**11-02-2026**

*Copyright © 1999 by Karl E. Wiegers. Permission is granted to use, modify, and distribute this document.*

# Table of Contents

**Table of Contents** ii
**Revision History** ii
**1. Introduction** 1
1.1 Purpose 1
1.2 Document Conventions 1
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 2
1.5 References 2
**2. Overall Description** 2
2.1 Product Perspective 2
2.2 Product Functions 2
2.3 User Classes and Characteristics 2
2.4 Operating Environment 2
2.5 Design and Implementation Constraints 2
2.6 User Documentation 2
2.7 Assumptions and Dependencies 3
**3. External Interface Requirements** 3
3.1 User Interfaces 3
3.2 Hardware Interfaces 3
3.3 Software Interfaces 3
3.4 Communications Interfaces 3
**4. System Features** 4
4.1 System Feature 1 4
4.2 System Feature 2 4
4.3 System Feature 3 4
4.4 System Feature 4 4
4.5 System Feature 5 4
**5. Other Nonfunctional Requirements** 4
5.1 Performance Requirements 4
5.2 Safety Requirements 5
5.3 Security Requirements 5
5.4 Software Quality Attributes 5
5.5 Business Rules 5
**6. Other Requirements** 5
**Appendix A: Glossary** 5
**Appendix B: Analysis Models** 5
**Appendix C: To Be Determined List** 6

# Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| | | | |

# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) document identifies the functions, features, and constraints of the BeeBuzz logistics platform. Its primary purpose is to provide a definitive technical and operational agreement between the development team and the project stakeholders. By detailing the interactions between "Flowers" (Shippers) and "Bees" (Drivers), this document ensures that all parties have a unified understanding of the system's architecture and performance expectations.

## 1.2 Document Conventions

To ensure clarity and scannability, this document follows these conventions:

- Bold text is used for UI elements, key terms, or specific requirement identifiers (e.g., REQ-1).
- Italicized text is used for internal personas or industry-specific jargon.
- **Prioritization**: Requirements are prioritized using the MoSCoW method (Must-have, Should-have, Could-have, Won't-have).
- **Standardization**: All technical requirements follow the IEEE 830-1998 standard for SRS documentation.

## 1.3 Intended Audience and Reading Suggestions

This document is intended for several distinct groups:

- **Project Managers & Stakeholders**: Should focus on the Product Scope and User Classes to ensure business goals are met.
- **Software Developers**: Should prioritize the Functional Requirements and External Interface sections for implementation details.
- **Quality Assurance (QA) Team**: Should use the Non-Functional Requirements to design test cases and acceptance criteria.
- **Legal & Compliance Officers**: Should review References and Security sections to ensure adherence to transportation laws and data privacy.

## 1.4 Product Scope

BeeBuzz is a digital freight-matching ecosystem designed to disrupt the traditional, manual brokerage model.

The software consists of a web-based portal for Shippers and a GPS-enabled mobile application for Truck Drivers. The system automates the lifecycle of a shipment—from posting a load and bidding to real-time tracking and automated payments.

**Key Benefits:**

- **For Shippers:** Lower overhead costs by bypassing traditional brokers and gaining direct access to verified carriers**.**
- **For Drivers:** Increased profitability by minimizing empty return trips (deadheading) and securing faster payouts.
- **For the Industry:** Improved transparency through a blockchain-inspired digital ledger of BOLs (Bills of Lading) and PODs (Proof of Delivery).

## 1.5 References

The development and compliance of BeeBuzz are guided by the following documents and standards:

1. **IEEE Std 830-1998**: Recommended Practice for Software Requirements Specifications.
2. **FMCSA Regulation**: Federal Motor Carrier Safety Administration guidelines for Electronic Logging Devices(ELD) and Hours of Services(HOS).
3. **PCI-DSS 4.0**: Data security standards for handling financial transactions and driver payouts.
4. **BeeBuzz Brand Identity Guidelines**: For UI/UX consistency across the “Bee” and “Flower” interfaces.

# 2. Overall Description

## 2.1 Product Perspective

BeeBuzz is a brand new mobile and web app that makes freight logistics way easier connecting shippers with goods to move and truck drivers ready to haul. It was born out of the mess in India's trucking world, where everyone relies on phone calls and WhatsApp chaos. This app swaps that for quick matching, live tracking, and smooth payments. It's standalone but buddies up with maps and payment tools.

## 2.2 Product Functions

For Shippers (Flowers):

- Register and verify account with phone/email
- Add company details and payment methods
- Post new loads with pickup/delivery locations, cargo type, weight, and truck requirements
- View all active and past shipments
- See list of driver bids with amounts and ratings
- Accept or reject driver bids
- Track assigned driver location in real-time on map
- Get estimated arrival time updates
- Receive proof of delivery photos and signatures
- Confirm delivery completion
- View payment history and invoices
- Rate drivers after delivery
- Chat with drivers directly in-app
- Get push notifications for bids and status updates

For Drivers (Bees):

- Register and verify account with phone
- Upload license, insurance, and vehicle documents
- Browse available loads on map or list
- Filter loads by location, type, weight, and payout
- Place bids on loads
- View bid status (pending/accepted/rejected)
- Get navigation to pickup and delivery points
- Update shipment status:
    - Arrived at Pickup
    - Loaded & Departed
    - En Route
    - Arrived at Delivery
    - Delivered
- Share live GPS location during active shipments
- Take photos of delivered goods as proof
- Upload signed documents or damage reports
- Receive automatic payments after delivery confirmation
- View earnings history
- Rate shippers after completing jobs
- Chat with shippers directly in-app
- Get push notifications for new loads and bid updates

System Functions (Automatic):

- Match loads to nearby drivers based on location and truck type
- Detect when driver enters pickup/delivery zone using GPS
- Send push notifications for important events
- Hold payments in escrow until delivery confirmed

- Verify driver documents before allowing loads
- Log all transactions and location history
- Flag disputes for admin review
- Generate invoices and receipts

## 2.3 User Classes and Characteristics

Three main folks we'll focus on:

- Shippers (top priority): Busy logistics managers who post loads daily. Comfortable with apps and require full tracking tools, bid management, and delivery confirmation features.
- Drivers (top priority): Truck operators frequently checking for available gigs. Have basic smartphone skills and need simple interfaces for bidding, navigation, and payment updates.
- Admins (support crew): Internal BeeBuzz team members who monitor platform activity, verify users, resolve disputes, and ensure smooth operations through admin dashboards.

## 2.4 Operating Environment

Runs on everyday phones Android 10+, iOS 14+, or web browsers. Powered by React Native apps, Node.js backend, and friends like Google Maps and Firebase. Handles spotty rural internet like a champ.

## *2.5 Design and Implementatio*n Constraints

Some practical ground rules:

- Built with React Native, MongoDB, and AWS.
- Phones need 2GB+ RAM; follow Indian privacy/GST rules.
- Security must-haves: 2FA logins, encrypted chats.
- English + Hindi, accessible design.

## 2.6 User Documentation

Easy guides included:

- PDF manuals for shippers/drivers/admins.
- In-app help and quick FAQs.
- Short video tutorials in English/Hindi.

## 2.7 Assumptions and Dependencies

What we're banking on: Users have decent GPS/internet, and local trucking spots catch on fast.
What we need: Stable Google Maps/Razorpay APIs, steady 4-week dev sprints.

# 3. External Interface Requirements

## 3.1 User Interfaces

The BeeBuzz application will have two different user interfaces: one for Shipper Mobile App users and another for Truck Driver Mobile App users. The application will be developed natively for iOS and Android platforms, incorporating the respective Human Interface Guidelines.

**Platform & Standards:**

- **IOS:** Adhere to Apple Human Interface Guidelines. Minimum target version: iOS 14.0.
- **Android:** Adhere to Google Material Design Guidelines. Minimum target version: API level 29 (Android 10).

**Shared UI Components:**

- **Login/Registration Screen:** A unified entry point will be used, with role-based navigation options for Shipper/Driver users. The screen will have features such as "Forget Password" and "Help”.
- **Bottom Navigation Bar:** The application will have core navigation options via a bottom tab navigation bar, which will be used on both Shipper Mobile App and Truck Driver Mobile App.
- **Standard Buttons:** Primary buttons such as "Post Load," "Bid," "Confirm Pickup," etc., will be of the same brand color with clear, action-oriented text. Secondary buttons will be outlined.
- **Error Messages:** In-app errors will be displayed via non-modal toast messages or inline validation, with clear, user-friendly error message text, such as "Invalid password. Please try again." Critical errors will be displayed via modal alerts.

- **Loading States:** Skeleton screens will be used to display loading states instead of blank screens or full-screen spinners.

**Shipper App Specifics:**

- **Dashboard:** Shows a list of active loads with their respective status (e.g., "Open for Bids," "Assigned," "In Transit"). A prominent "+" button to start the "Post a Load" feature.
- **Post a Load Screen:** A form that consists of multiple sections for entering details such as origin/destination (with geolocation feature), type of cargo, weight, type of truck needed (e.g., "Refrigerated," "Flatbed"), and preferred time slots for pickup/delivery. Input validation for each field.
- **Bid Management Screen:** A list view of drivers bidding for a load, including their respective bid amount, rating, and estimated time of arrival. Tapping a bid item would open a new screen to accept/reject a bid.
- **Tracking Screen:** A full-screen map view (Google/Apple Maps SDK integration) showing the driver's current location, route, and estimated time of arrival.

**Driver App Overview:**

- **Dashboard:** View all available loads on the map and in a list. Filter loads by load type, search within a certain radius, and sort by payout range.
- **Load Details:** All the details about the load are visible at a glance, including the pickup location, destination, distance, weight, cargo, and the price offered. The large, attention-grabbing “Place Bid” button is also visible.
- **Bid/Job Status:** Keep track of the status of your bids, whether accepted or rejected, and your active jobs. For active jobs, large buttons are available to update the status, including “Arrived at Pickup,” “Loaded & Departed,” “Arrived at Delivery,” “Delivered,” etc.
- **In-App Navigation:** When you start your trip, the driver app opens the native navigation app on your phone, Google Maps or Apple Maps, with the destination entered automatically.

**User Interface:** The complete screen flow, wireframes, and final mock-ups for the entire application are described in the BeeBuzz UI/UX Design Specification Document.

## 3.2 Hardware Interfaces

The BeeBuzz mobile apps are built to integrate well with the hardware the device is running on, utilizing what is available to it to perform the required functions.

**GPS Receiver:**

- **Requirement:** The Driver App must access the GPS capabilities built into the phone or tablet to obtain the driver’s location.
- **Purpose:** This data is used to display the driver's position on a map for the shipper, calculate ETAs, and verify geofenced arrivals at pickup and delivery locations.

- **Control:** The app will require the necessary permissions to access the location on the phone or tablet, including background location access when the driver has active loads, as well as foreground location access for general browsing.

**Camera:**

- **Requirement:** The app will utilize the camera on the phone or tablet.
- **Purpose:**
    - **Drivers:** the camera will be used to upload proof-of-delivery documents, such as photos of the signed bill of lading, damaged cargo, or proof of delivery.
    - **Shippers:** the camera will be used to upload photos or other information regarding cargo, such as special handling requirements, when creating the load.

**Cellular/Wi-Fi Radio:**

- **Requirement:** The app requires access to the internet via the phone or tablet's cellular or Wi-Fi radio to perform all the required functions.
- **Data:** All data is transferred over this connection.

## 3.3 Software Interfaces

BeeBuzz interfaces with multiple external software pieces and backend systems, and in all cases, the interaction is initiated from the mobile application.

- **Backend API Server (BeeBuzz Core)**
    - **Name & Version:** Custom BeeBuzz API / v1
    - **Nature of Communication:** All data exchanges between the mobile applications and the BeeBuzz platform are facilitated via this RESTful API.
    - **Protocol:** HTTPS using HTTP/1.1 or HTTP/2, using JSON as the data exchange format.
    - **Authentication:** OAuth2 tokens are used for authenticating the API requests. After authenticating, the application will obtain an access token that is required to be sent in the Authorization header in subsequent requests.
    - **Services & Data:**
        - **User Service:** Login, registration, and profile management are facilitated via the User Service, which exposes endpoints such as /users, /login, POST/PUT /users.
        - **Load Service:** Shippers create loads via POST /loads, and drivers retrieve the available loads via GET /loads?status=open.
        - **Bidding Service:** Drivers can place their bids via POST /bids, and shippers can view and accept the bids via GET/PUT /bids/{id}.
        - **Tracking Service:** Drivers send their GPS coordinates periodically via POST /tracking, and shippers can retrieve the latest GPS coordinates via GET /tracking/{loadId}.

- **Maps & Geocoding Service**
  - **Name & Version:** Google Maps SDK for iOS and Android / [Latest Stable Version]
  - **Purpose:** Map rendering, location display, and place autocomplete.
  - **Nature:** The application uses the native SDK with direct access to Google's servers. It deals with map tile rendering and UI widgets. Geocoding is handled by passing the address string to the SDK to obtain the coordinates.
- **Cloud Notification Service:**
  - **Name & Version:** Firebase Cloud Messaging (FCM) for Android / Apple Push Notification Service (APNs) for iOS
  - **Purpose:** Send push notifications from the BeeBuzz backend to the mobile application.
  - **Data:** The notification contains a payload with message type (NEW_BID, LOAD_ACCEPTED, DOCUMENT_REQUIRED), etc., along with some minimal data set like load_id: '12345' to open the appropriate screen on tap.
- **Database:**
  - **Name & Version:** PostgreSQL / 14.x (Backend), SQLite / 3.x (Mobile Local Storage)
  - **Nature:**
    - **Backend:** The BeeBuzz API server writes to the primary PostgreSQL database to store user, load, bid, transactional data, etc.
    - **Mobile (Local):** The application uses a local SQLite database to store some of the important data in the mobile application to view locally in offline mode.

## 3.4 Communications Interfaces

BeeBuzz is entirely internet-based for both live and deferred operations, and security and reliability are our highest priorities.

- **Primary Communication Protocol:** All data exchanged between apps is done over HTTPS (HTTP over TLS 1.2 and later). This ensures all bits of information exchanged between our mobile application and our backend are always encrypted and cannot be intercepted.
- **Real-time Tracking:**
  - **Mechanism:** Driver updates are sent to our backend at regular intervals (e.g., every 30 seconds) or upon a significant change in their position via HTTPS POST to our tracking API endpoint.
  - **Shipper Updates:** The Shipper application periodically polls our tracking API (approximately every 15 seconds) to retrieve the latest driver position and updated ETA while viewing an active load.
- **Push Notifications:**
  - **Protocol:** Asynchronous messages via FCM/APNs as specified in Section 3.3.
  - **Purpose:** Send time-sensitive messages to users without the need to have our application open to notify users of new bids, bid status updates, etc.

- **Security & Encryption:**
  - All data transmitted via HTTPS will be encrypted.
  - Data sent in push notifications will not include sensitive user data, such as full names and actual bid amounts, to ensure user data is kept private in case the notification is displayed when the device is locked.
- **Data Rates & Synchronization:**
  - The applications are designed to use minimal data, such as compressing images before they are sent and using the API to send only necessary JSON data in the response.
  - For critical user actions, such as when a driver changes their status to "Arrived at Pickup," the application will store the data locally in case the network is unavailable and will sync the data once the network is available again.

# 4. System Features

## 4.1 Shipper Account and Profile Management

### 4.1.1 Description and Priority

This feature enables shippers (Flowers) to create and manage their accounts, verify their identity, and configure company details and payment methods. It serves as the foundation for all shipper interactions with the platform.

Priority: High

### 4.1.2 Stimulus/Response Sequences

Stimulus:
- New user downloads the BeeBuzz Shipper app and selects "Register"
- User enters phone number or email for verification
- System sends verification code
- User enters verification code
- User completes company profile and adds payment method

Response:
- System validates and verifies the account
- System creates secure user profile
- System stores encrypted payment information
- System displays welcome dashboard with available functions

### 4.1.3 Functional Requirements

REQ-SHI-REG-1: The system shall allow new shippers to register using either a valid phone number (with country code) or email address.

REQ-SHI-REG-2: The system shall send a one-time password (OTP) to the provided phone number or email for verification before account activation.

REQ-SHI-REG-3: The system shall require OTP entry within a configurable time window (default: 5 minutes) and invalidate expired codes.

REQ-SHI-REG-4: The system shall prevent duplicate registrations using the same phone number or email.

REQ-SHI-PRO-1: The system shall allow shippers to add and edit company details including:
- Company name and legal business name
- GSTIN or tax identification number
- Business address
- Contact phone and email
- Company logo (optional)

REQ-SHI-PRO-2: The system shall validate GSTIN format according to government standards before acceptance.

REQ-SHI-PAY-1: The system shall allow shippers to add multiple payment methods including:
- Credit/debit cards
- Bank account details for direct transfers
- UPI IDs
- Company credit account (if applicable)

REQ-SHI-PAY-2: The system shall encrypt all payment method data following PCI-DSS compliance standards.

REQ-SHI-PAY-3: The system shall allow shippers to set a default payment method and edit/remove existing methods.

## 4.2 Load Posting and Management

### 4.2.1 Description and Priority

This feature allows shippers to create new shipment requests (loads) with comprehensive details about pickup/delivery locations, cargo specifications, and truck requirements. Shippers can also view and manage all their active and past shipments.

Priority: High

### 4.2.2 Stimulus/Response Sequences

Stimulus:
- Shipper navigates to "Post New Load" screen
- Shipper enters pickup location (address or map pin)
- Shipper enters delivery location (address or map pin)

- Shipper specifies cargo type, weight, dimensions
- Shipper selects required truck type and special requirements
- Shipper sets preferred pickup and delivery time windows
- Shipper submits the load for posting

Response:
- System validates all required fields
- System calculates estimated distance and route
- System publishes load to available drivers
- System adds load to shipper's active shipments list
- System sends confirmation to shipper

### 4.2.3 Functional Requirements

REQ-SHI-LOAD-1: The system shall provide a multi-step form interface for creating new loads with clear field labels and validation.

REQ-SHI-LOAD-2: The system shall integrate with mapping services to autocomplete addresses and capture precise GPS coordinates for pickup and delivery locations.

REQ-SHI-LOAD-3: The system shall require the following fields before allowing load publication:
- Pickup location (address and/or map coordinates)
- Delivery location (address and/or map coordinates)
- Cargo type (from predefined list with "other" option)
- Cargo weight (in kg or tons)
- Required truck type (from predefined list: container, flatbed, refrigerated, etc.)
- Pickup date and preferred time window
- Delivery date and preferred time window

REQ-SHI-LOAD-4: The system shall allow optional fields including:
- Cargo dimensions (length, width, height)
- Special handling instructions
- Hazardous materials declaration
- Reference numbers or purchase order IDs

REQ-SHI-LOAD-5: The system shall calculate and display estimated distance and route preview before load publication.

REQ-SHI-LIST-1: The system shall provide a dashboard view showing all active loads with status indicators (Open for Bids, Assigned, In Transit, Delivered).

REQ-SHI-LIST-2: The system shall allow shippers to filter and search past shipments by date range, status, or reference number.

## 4.3 Bid Management and Driver Selection

### 4.3.1 Description and Priority

This feature enables shippers to view driver bids on their posted loads, evaluate drivers based on bid amounts and ratings, and make acceptance or rejection decisions.

Priority: High

### 4.3.2 Stimulus/Response Sequences

Stimulus:
- Drivers place bids on a shipper's posted load
- System sends push notification to shipper
- Shipper opens Bid Management screen for the load
- Shipper reviews list of driver bids with amounts and ratings
- Shipper selects a bid to accept (or rejects specific bids)
- Shipper confirms selection

Response:
- System locks the load to the selected driver
- System notifies all bidders of the decision (accepted/rejected)
- System updates load status to "Assigned"
- System makes load details available to the assigned driver

### 4.3.3 Functional Requirements

REQ-SHI-BID-1: The system shall display all bids received for a specific load in a sorted list (default: by bid amount lowest to highest).

REQ-SHI-BID-2: For each bid, the system shall display:
- Driver's name (or username/handle)
- Bid amount
- Driver's average rating (stars and numerical)
- Number of completed jobs
- Estimated time to reach pickup location
- Vehicle type and capacity

REQ-SHI-BID-3: The system shall allow shippers to view detailed driver profiles including complete rating history and completed job counts.

REQ-SHI-BID-4: The system shall allow shippers to accept exactly one bid per load, which automatically rejects all other pending bids.

REQ-SHI-BID-5: The system shall provide a confirmation dialog before finalizing bid acceptance to prevent accidental selections.

REQ-SHI-BID-6: The system shall notify all drivers via push notification when their bid is accepted or rejected.

## 4.4 Real-Time Tracking and Monitoring

### 4.4.1 Description and Priority

This feature provides shippers with real-time visibility into their assigned driver's location, estimated arrival times, and shipment progress throughout the delivery journey.

Priority: High

### 4.4.2 Stimulus/Response Sequences

Stimulus:
- Shipper opens Tracking screen for an active shipment
- System displays map with driver's current location
- Driver updates shipment status (e.g., "Loaded & Departed")
- System updates shipper with new status and ETA
- Driver's GPS location updates continuously during transit

Response:
- Shipper sees real-time driver location on map
- Shipper receives status change notifications
- System recalculates and displays updated ETA
- Shipper can monitor progress until delivery

### 4.4.3 Functional Requirements

REQ-SHI-TRK-1: The system shall provide a full-screen map view showing the assigned driver's real-time location during active shipments.

REQ-SHI-TRK-2: The map shall display:
- Driver's current position with moving icon
- Pickup location marker
- Delivery location marker
- Planned route (if available)

REQ-SHI-TRK-3: The system shall update driver location at intervals not exceeding 10 seconds when the shipment is active.

REQ-SHI-TRK-4: The system shall display the current shipment status prominently on the tracking screen.

REQ-SHI-TRK-5: The system shall calculate and display estimated time of arrival at the delivery location, updated whenever driver location updates.

REQ-SHI-TRK-6: The system shall generate push notifications for shippers when:
- Driver marks "Arrived at Pickup"
- Driver marks "Loaded & Departed"
- Driver marks "Arrived at Delivery"
- Driver marks "Delivered"

## 4.5 Delivery Confirmation and Proof of Delivery

### 4.5.1 Description and Priority

This feature enables shippers to receive and review proof of delivery documentation, including photos and signatures, and formally confirm delivery completion.

Priority: High

### 4.5.2 Stimulus/Response Sequences

Stimulus:
- Driver arrives at delivery location
- Driver takes photos of delivered goods
- Driver captures recipient signature (if applicable)
- Driver uploads documents and marks "Delivered"
- System notifies shipper of delivery completion
- Shipper reviews proof of delivery documents

Response:
- System displays proof of delivery to shipper
- System prompts shipper to confirm delivery
- Shipper confirms acceptance (or flags issues)
- System initiates payment processing upon confirmation

### 4.5.3 Functional Requirements

REQ-SHI-POD-1: The system shall notify shippers immediately when a driver marks a shipment as "Delivered" and uploads proof of delivery.

REQ-SHI-POD-2: The system shall display all proof of delivery documents in a gallery view, including:
- Photos of delivered goods
- Captured signature images
- Uploaded PDFs or document scans
- Delivery timestamp and GPS coordinates

REQ-SHI-POD-3: The system shall allow shippers to confirm delivery completion with a single tap/click after reviewing documents.

REQ-SHI-POD-4: The system shall provide an option for shippers to flag discrepancies or issues with the delivery, including:
- Missing or damaged items
- Incorrect delivery location
- Incomplete documentation

REQ-SHI-POD-5: When issues are flagged, the system shall:
- Prevent automatic payment release
- Notify the driver of the issue
- Flag the shipment for admin review

## 4.6 Driver Account and Document Verification

### 4.6.1 Description and Priority

This feature enables drivers (Bees) to create accounts, verify their identity, and upload required documentation (license, insurance, vehicle documents) to establish trust and compliance before they can accept loads.

Priority: High

### 4.6.2 Stimulus/Response Sequences

Stimulus:
- New driver downloads BeeBuzz Driver app and selects "Register"
- Driver enters phone number for verification
- System sends OTP verification code
- Driver enters code to verify account
- Driver completes profile and uploads required documents
- System verifies documents (automated + manual review)

Response:
- Account created with "Pending Verification" status
- Documents stored securely for review
- Driver notified when verification is complete
- Verified driver can now browse and bid on loads

### 4.6.3 Functional Requirements

REQ-DRV-REG-1: The system shall allow new drivers to register using a valid phone number with country code.

REQ-DRV-REG-2: The system shall send a one-time password (OTP) via SMS for phone verification before account activation.

REQ-DRV-REG-3: The system shall prevent multiple registrations using the same phone number.

REQ-DRV-DOC-1: The system shall require drivers to upload the following documents before they can bid on loads:
- Driver's license (front and back)
- Commercial vehicle insurance certificate
- Vehicle registration certificate (RC)
- Vehicle permit (if applicable)
- Driver's photo for profile

REQ-DRV-DOC-2: The system shall support document upload via camera capture or gallery/file selection.

REQ-DRV-DOC-3: The system shall validate document images for readability, expiration dates, and document type matching.

REQ-DRV-DOC-4: The system shall maintain document verification status (Pending, Verified, Rejected) and notify drivers of changes.

REQ-DRV-DOC-5: The system shall prevent drivers with unverified or expired documents from bidding on or accepting loads.

## 4.7 Load Browsing and Bidding

### 4.7.1 Description and Priority

This feature enables drivers to discover available loads through map and list views, apply filters, and place competitive bids on loads that match their preferences and qualifications.

Priority: High

### 4.7.2 Stimulus/Response Sequences

Stimulus:
- Verified driver opens the Driver app
- System displays available loads on map and in list
- Driver applies filters to narrow results
- Driver selects a load and places a bid
- System processes bid and updates bid status

Response:
- Available loads displayed with key information
- Filtered results update in real-time
- Bid is recorded and visible to shipper
- Bid appears in driver's "My Bids" list with "Pending" status

### 4.7.3 Functional Requirements

REQ-DRV-BRW-1: The system shall provide a default view showing available loads on an interactive map with load markers.

REQ-DRV-BRW-2: The system shall also provide a list view showing available loads in a sortable, scrollable list.

REQ-DRV-BRW-3: Both views shall display key load information:
- Pickup and delivery locations (city/area)
- Distance (estimated)
- Cargo type and weight
- Payout amount
- Pickup time window

REQ-DRV-BRW-4: The system shall provide filtering options by location radius, cargo type, weight range, payout range, and required truck type.

REQ-DRV-BID-1: The system shall allow verified drivers to place bids on available loads that match their vehicle type and document status.

REQ-DRV-BID-2: The bid submission screen shall display load summary, current lowest bid (if visible), and fields for bid amount and optional notes.

REQ-DRV-BID-3: The system shall provide a "My Bids" screen showing all bids placed by the driver with status (Pending, Accepted, Rejected).

REQ-DRV-BID-4: The system shall send push notifications for bid status changes (accepted, rejected, expired).

## 4.8 Trip Execution and Proof of Delivery

### 4.8.1 Description and Priority

This feature guides drivers through the complete trip lifecycle with predefined status updates, GPS location sharing, and proof of delivery capture to ensure successful completion.

Priority: High

### 4.8.2 Stimulus/Response Sequences

Stimulus:
- Driver accepts/gets assigned to a load
- Driver navigates to pickup location and marks "Arrived at Pickup"
- Driver loads goods and marks "Loaded & Departed"
- Driver navigates to delivery location and marks "Arrived at Delivery"
- Driver captures proof of delivery and marks "Delivered"

Response:
- System records each status change with timestamp and GPS
- Shipper receives status update notifications
- System shares live location during active transit
- System stores proof of delivery and notifies shipper

### 4.8.3 Functional Requirements

REQ-DRV-TRIP-1: The system shall display all active jobs on the driver's dashboard with clear status indicators.

REQ-DRV-TRIP-2: For each active job, the system shall provide pickup and delivery locations, shipper contact information, and cargo details.

REQ-DRV-TRIP-3: The system shall provide a prominent status update button that cycles through predefined statuses:
- Arrived at Pickup

- Loaded & Departed
- En Route
- Arrived at Delivery
- Delivered

REQ-DRV-TRIP-4: The system shall integrate with device navigation apps with one-tap launch to navigate to pickup and delivery addresses.

REQ-DRV-TRIP-5: The system shall automatically share live GPS location during active shipments.

REQ-DRV-POD-1: The system shall provide a dedicated "Proof of Delivery" screen for capturing photos and recipient signatures.

REQ-DRV-POD-2: The system shall support multiple photo capture and signature capture directly on the device screen.

REQ-DRV-POD-3: The system shall automatically attach GPS coordinates and timestamp to all proof of delivery submissions.

## 4.9 Automated Payments and Earnings

### 4.9.1 Description and Priority

This feature automatically processes payments to drivers upon successful delivery confirmation and provides comprehensive earnings history and transaction records.

Priority: High

### 4.9.2 Stimulus/Response Sequences

Stimulus:
- Shipper confirms delivery completion
- System initiates automatic payment to driver
- Driver receives payment notification
- Driver opens "Earnings" section to view balance and history

Response:
- Payment processed and transferred to driver's account
- Transaction added to earnings history
- Driver can view detailed breakdown of earnings
- Driver can access payment summaries and receipts

### 4.9.3 Functional Requirements

REQ-DRV-PAY-1: The system shall automatically process payment to drivers upon successful delivery confirmation by the shipper.

REQ-DRV-PAY-2: The system shall display a current earnings summary showing total earnings, pending payments, and available balance.

REQ-DRV-PAY-3: The earnings history view shall include for each transaction:
- Load reference and details
- Shipper name
- Delivery date
- Gross earnings
- Platform fees (if applicable)
- Net amount paid
- Payment status

REQ-DRV-PAY-4: The system shall generate payment receipts for each completed transaction, accessible from the earnings history.

REQ-DRV-PAY-5: The system shall notify drivers via push notification when payment is processed or requires action.

REQ-DRV-PAY-6: The system shall support multiple payout methods (bank transfer, UPI, digital wallet) based on driver preference.

## 4.10 Escrow Payment System

### 4.10.1 Description and Priority

This system feature securely holds shipper payments in escrow when a load is accepted, releasing funds to drivers only after successful delivery confirmation, protecting both parties.

Priority: High

### 4.10.2 Stimulus/Response Sequences

Stimulus:
- Shipper accepts driver's bid on a load
- System calculates total amount (bid + fees)
- System places hold on shipper's payment method for the amount
- Funds held in escrow until delivery confirmation
- Shipper confirms successful delivery
- System releases payment to driver minus platform fees

Response:
- Payment hold confirmed to both parties
- Shipper notified of hold placement
- Driver notified of payment guarantee
- Upon delivery confirmation, payment processed to driver
- Transaction records updated for both parties

### 4.10.3 Functional Requirements

REQ-SYS-ESC-1: The system shall place a hold on the shipper's selected payment method for the full load amount (including fees) at the time of driver acceptance.

REQ-SYS-ESC-2: The system shall clearly communicate to both parties that funds are held in escrow and guaranteed to the driver upon successful delivery.

REQ-SYS-ESC-3: The system shall only release funds from escrow when:
- Shipper confirms delivery completion, OR
- 24 hours have passed since driver marked "Delivered" without shipper dispute (automatic release), OR
- Admin resolves dispute in driver's favor

REQ-SYS-ESC-4: The system shall return funds to shipper if:
- Load is cancelled before pickup (with both parties consent)
- Admin resolves dispute in shipper's favor
- Driver fails to complete pickup within agreed timeframe

REQ-SYS-ESC-5: The system shall calculate and deduct platform fees before releasing the remaining amount to the driver.

REQ-SYS-ESC-6: All escrow transactions shall be logged with a complete audit trail including timestamps, parties involved, and amounts.

REQ-SYS-ESC-7: The system shall notify both parties of all escrow status changes (hold placed, released, refunded).

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

We want BeeBuzz to feel snappy and reliable, even when everyone's rushing to book trucks during peak seasons like festivals. This keeps shippers happy and drivers on the road without frustration.

- Everyday actions like booking a truck or checking status should wrap up in under 2 seconds for 95% of users when things are busy (say, 1,000 people online at once).
- It needs to handle up to 10,000 users without slowing down more than 5%, thanks to cloud scaling.
- Real-time stuff, like GPS tracking or alerts, refreshes every 10 seconds for ongoing jobs.
- Why? Logistics can't afford delays, think fresh produce spoiling or urgent deliveries missing deadlines.

## 5.2 Safety Requirements

Safety comes first because lives and cargo are on the line. We're following India's Motor Vehicles Act and road safety rules to keep everyone protected.

- Drivers hit "all good" before taking a load; the app blocks them if it spots fatigue via phone sensors.
- Easy one-tap reporting for accidents with photos/videos, instantly pinging shippers and cops.
- Smart alerts for overloads based on truck details, plus no-go zones for risky roads.
- Goal: Meet ISO 39001 standards for road safety management.

## 5.3 Security Requirements

Nobody wants their shipment details or location leaked. We're locking this down per India's DPDP Act to build trust.

- Logins need OTP plus fingerprint/face scan; shippers only see their own stuff.
- Everything GPS, payments, storage gets top-tier encryption (AES-256, PCI-DSS for cards).
- Drivers control location sharing; we log every access for audits.
- Aiming for ISO 27001 and routine security checks.

## 5.4 Software Quality Attributes

These are the "nice-to-haves" that make BeeBuzz shine for users (especially drivers in remote areas) and are easy for us to tweak later.

| Attribute | What We Want | How We'll Check |
|---|---|---|
| Availability | Up 99.9% of the time | Track with uptime monitors |
| Usability | Super easy, even for first-timers | 90% nail tasks on try #1 in tests |
| Reliability | No lost data if it crashes | Backups every 5 mins, auto-tested |
| Maintainability | Code that's a breeze to update | Keep changes under 10% per release |
| Portability | Works on phones, web, anywhere | Test across Android/iOS/browsers |
| Priority | Easy to use beats fancy portability | Team sign-off |

### 5.5 Business Rules

Simple ground rules to keep things fair and smooth, no new features, just enforcers.

- Only licensed, insured drivers with matching trucks get gigs.
- No sneaky changes to accepted jobs without driver OK.
- Payments drop after delivery proof (GPS + pic).
- Disputes? Try sorting it yourselves for 24 hours first, then admins step in.

## 6. Other Requirements

A few extras to round it out:

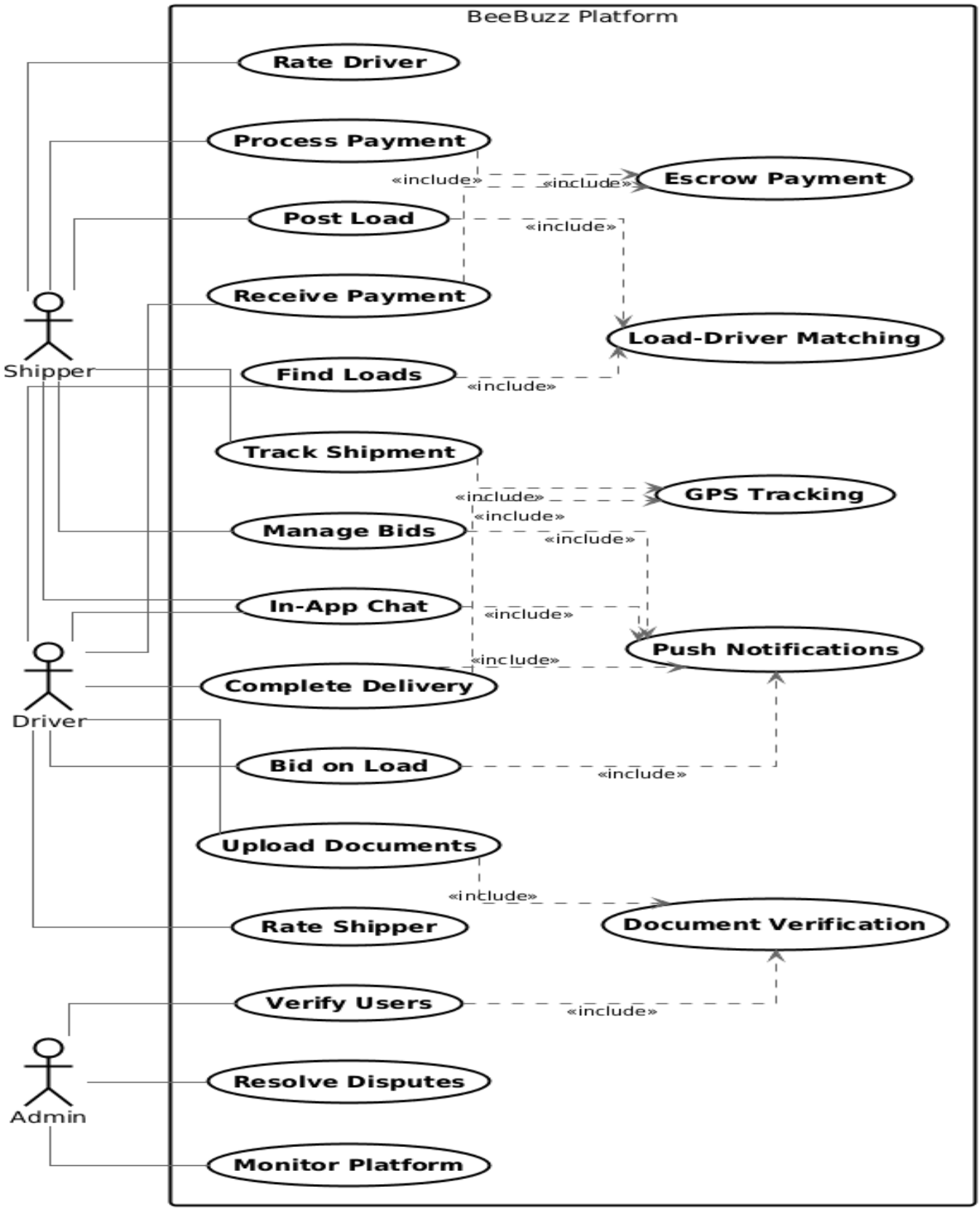
- Database: PostgreSQL for rock-solid transactions, MongoDB for live logs geared for 1M+ shipments yearly.
- Languages: Start with English/Hindi (Tamil too, for spots like Puducherry); ready for more.
- Legal: Built-in GST billing; all data stays in India (e.g., AWS Mumbai).
- Integrations: Open APIs for warehouses or partners.
- Accessibility: Voice features for hands-free driving, hitting WCAG 2.1 AA.

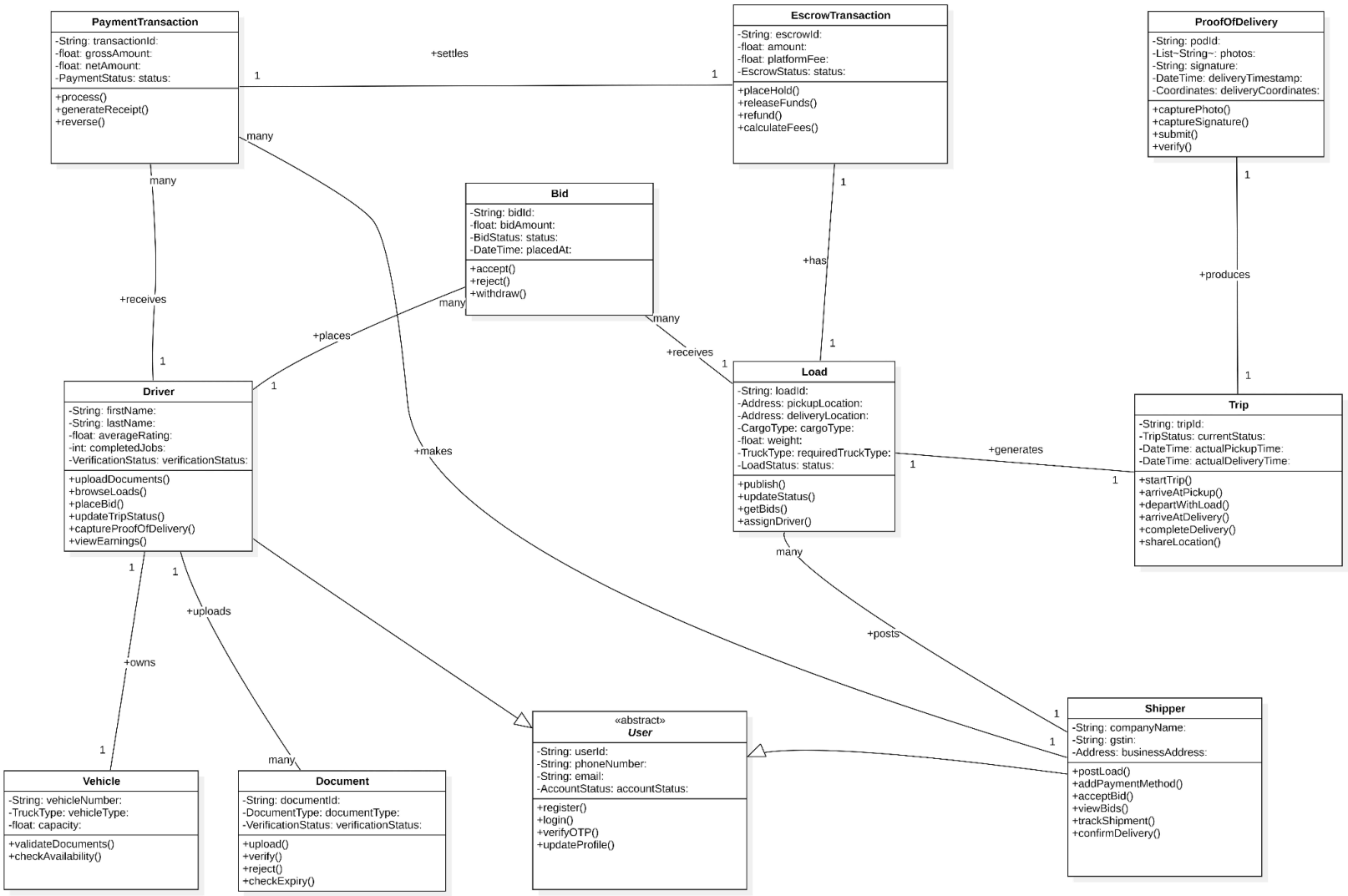
## Appendix A: Glossary

BeeBuzz speak:

- Shipper/Flower: Load poster.
- Driver/Bee: Job hauler.
- Load/Bid/ETA: Freight offer/arrival.
- Escrow/BOL/POD: Safe pay/proof docs.
- Geofence/OTP: GPS trigger/login code.

## Appendix B: Analysis Models


BeeBuzz Platform
Rate Driver
Process Payment
Escrow Payment
Post Load
Receive Payment
Load-Driver Matching
Find Loads
Track Shipment
GPS Tracking
Manage Bids
In-App Chat
Push Notifications
Complete Delivery
Bid on Load
Upload Documents
Rate Shipper
Document Verification
Verify Users
Resolve Disputes
Monitor Platform
Shipper
Driver
Admin
«include»




# Appendix C: To Be Determined List

1. Payment gateway fees.
2. Free tier limits.
3. Language rollout dates.
4. GPS battery goal.
5. Admin designs.
6. GST legal OK.